E. Sending a Sequence Over the Network

time limit per test: 1 second
memory limit per test: 256 megabytes

The sequence a is sent over the network as follows:

1. sequence a is split into segments (each element of the sequence belongs to exactly one segment, each segment is a group of consecutive elements of sequence);
2. for each segment, its length is written next to it, either to the left of it or to the right of it;
3. the resulting sequence b is sent over the network.

For example, we needed to send the sequence $a = [1, 2, 3, 1, 2, 3]$. Suppose it was split into segments as follows: $[{\color{red}1}] + [{\color{blue}2, 3, 1}] + [{\color{green}2, 3}]$. Then we could have the following sequences:

- $b = [1, {\color{red}1}, 3, {\color{blue}2, 3, 1}, {\color{green}2, 3}, 2]$,
- $b = [{\color{red}1}, 1, 3, {\color{blue}2, 3, 1}, 2, {\color{green}2, 3}]$,
- $b = [{\color{red}1}, 1, {\color{blue}2, 3, 1}, 3, 2, {\color{green}2, 3}]$,
- $b = [{\color{red}1}, 1, {\color{blue}2, 3, 1}, 3, {\color{green}2, 3}, 2]$.

If a different segmentation had been used, the sent sequence might have been different.

The sequence b is given. Could the sequence b be sent over the network? In other words, is there such a sequence a that converting a to send it over the network could result in a sequence b?

Input

The first line of input data contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

Each test case consists of two lines.

The first line of the test case contains an integer n ($1 \le n \le 2 \cdot 10^5$) — the size of the sequence b.

The second line of test case contains n integers $b_1, b_2, \dots, b_n$ ($1 \le b_i \le 10^9$) — the sequence b itself.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case print on a separate line:

- YES if sequence b could be sent over the network, that is, if sequence b could be obtained from some sequence a to send a over the network.
- NO otherwise.

You can output YES and NO in any case (for example, strings yEs, yes, Yes and YES will be recognized as positive response).

Example

input
7
9
1 1 2 3 1 3 2 2 3
5
12 1 2 7 5
6
5 7 8 9 10 3
4
4 8 6 2
2
3 1
10
4 6 2 1 9 4 9 3 4 2
1
1

output
YES
YES
YES
NO
YES
YES
NO

Note

In the first case, the sequence b could be obtained from the sequence $a = [1, 2, 3, 1, 2, 3]$ with the following partition: $[1] + [2, 3, 1] + [2, 3]$. The sequence b: $[1, 1, 2, 3, 1, 3, 2, 2, 3]$.

In the second case, the sequence b could be obtained from the sequence $a = [12, 7, 5]$ with the following partition: $[12] + [7, 5]$. The sequence b: $[12, 1, 2, 7, 5]$.

In the third case, the sequence b could be obtained from the sequence $a = [7, 8, 9, 10, 3]$ with the following partition: $[7, 8, 9, 10, 3]$. The sequence b: $[5, 7, 8, 9, 10, 3]$.

In the fourth case, there is no sequence a such that changing a for transmission over the network could produce a sequence b.